Case 13: Where does new work come from?

New specialties, technological change, and the demand for labour

1 Context

Continuous technological advancement changes the nature of work, continuously taking over some of the tasks performed by workers and introducing new tasks for workers to perform. An automatic mail sorter can take over work previously done by postal employees. A new solar-power system can create work for people who design, install, and maintain it. Both are technological advances, but they change the demand for human work in different ways. One makes an existing activity possible with less labour; the other introduces capabilities that can require new expertise.

New work need not involve a completely new occupation. An electrician who specializes in solar installations builds on an existing trade. A cybersecurity analyst develops a specialty within computing. New work can also emerge without an obvious technological breakthrough: an expanding market for personal care can support new specialties in services that people have purchased for many years.

Breakthroughs in robotics and artificial intelligence have fired alarms about the role of technology in taking over tasks, but we should also examine the role twin role of new job creation. What generates new activities for workers, who performs them, and how do they affect employment? Evidence from the United States between 1940 and 2018 makes it possible to examine these questions over several generations of technological change.

2 How can we identify new work?

The key to examine new work across the whole economy is the availability of job information in national censuses. When people complete a census, they describe their jobs in their own words. Census staff use an alphabetical index of detailed job titles to assign these responses to broader occupations. For example, a specialized computing title can be classified within computer scientists and systems analysts. As unfamiliar titles appear repeatedly, staff review whether they represent sufficiently distinct and prevalent activities to be added to the index.

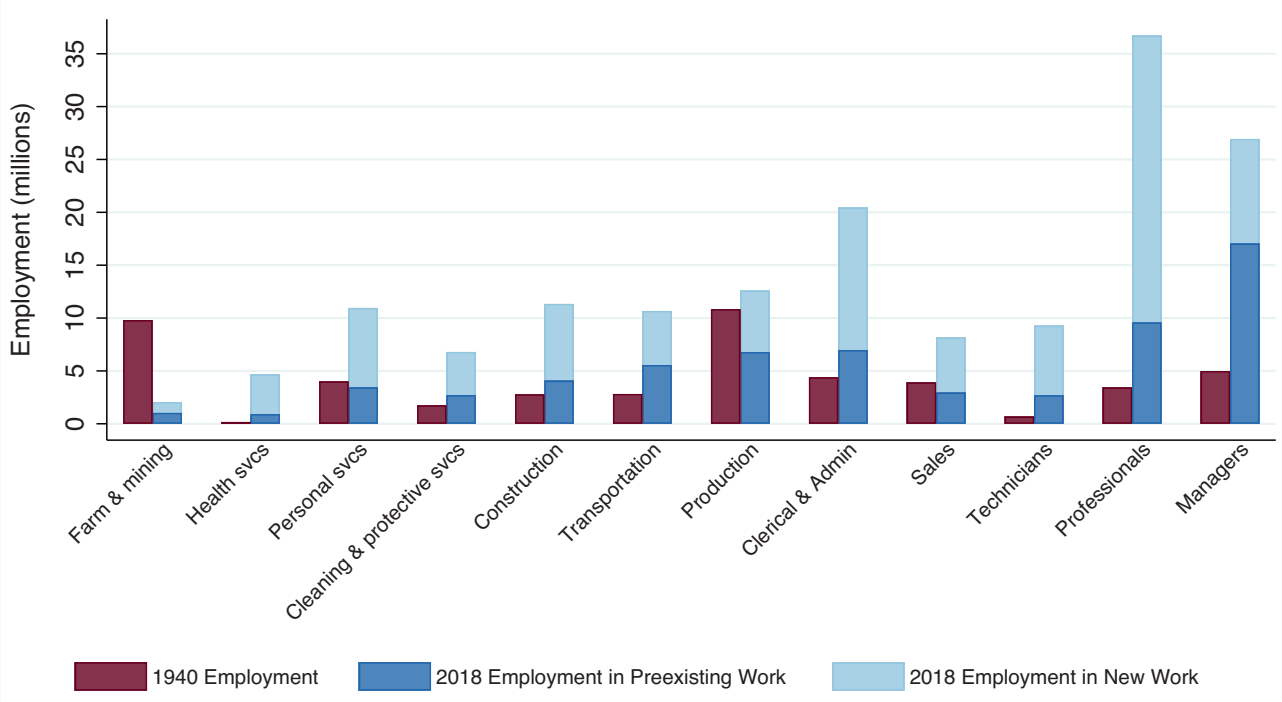
In this way, comparing successive editions of census data identifies newly recognized specialties. The index added *airplane designer* in 1950, *mental-health counselor* in 1970, *artificial intelligence specialist* in 2000, *wind-turbine technician* in 2010, and *cybersecurity analyst* in 2018. These detailed titles provide a measure of the emergence of new work.

We can also use the census information to get an estimate of how many people hold new specialties. To estimate that number, multiply employment in a broad occupation by the fraction of its detailed titles that are new:

$$\widehat{E}_o^{\text{new}} = E_o \frac{\text{new detailed titles in occupation } o}{\text{all detailed titles in occupation } o}. \tag{1}$$

This assigns the new titles their proportional share of the occupation's employment.

Exhibit 1. Employment in old and new specialties



Historical evidence. The vertical axis is employment in millions. For each broad occupation, the red bar shows 1940 employment. The stacked blue bar shows 2018 employment, divided into titles already present in 1940 and titles added subsequently. Broad occupations are ordered from lower to higher pay.

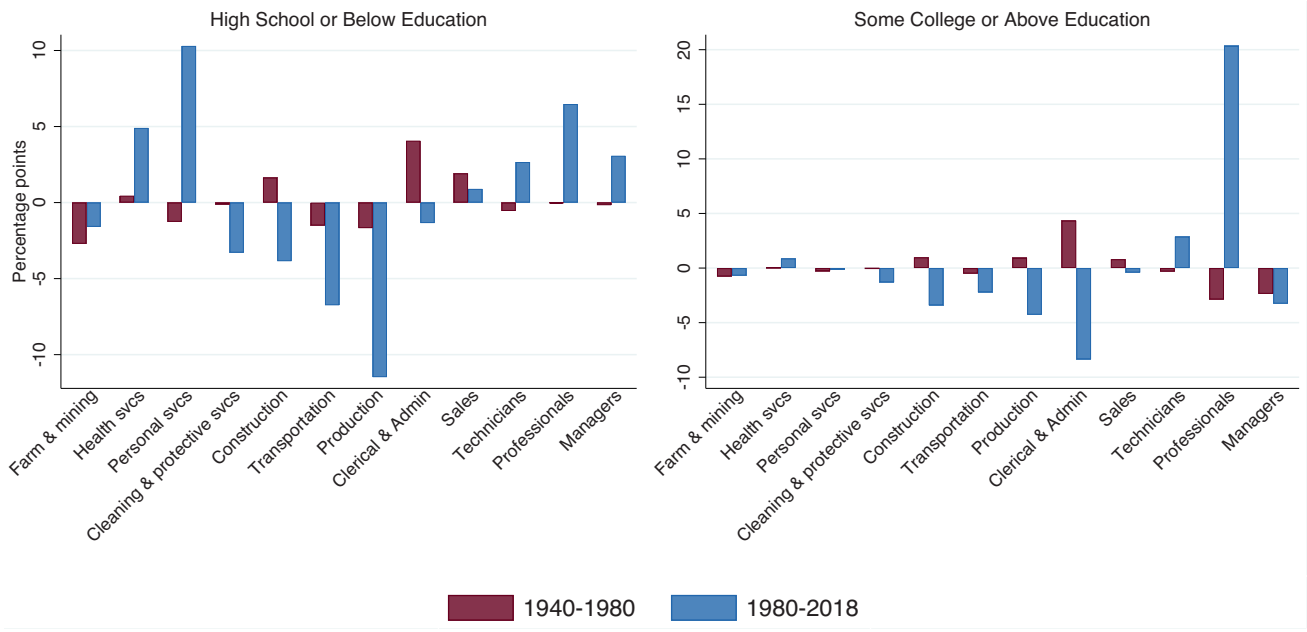
Approximately 60% of employment in 2018 is estimated to be in titles introduced after 1940. The estimated share is 74% among professionals and 46% in production occupations. The pattern shows substantial renewal within broad occupations, including occupations whose total employment grew relatively little.

3 Who performs the new work?

New work does not appear evenly across the occupational distribution. Exhibit 2 compares the distribution of new work with the distribution of existing work, separately for two education groups and two periods. Each bar is the occupation's share of employment in new work minus its share in existing work within the relevant group and period. A positive bar means the occupation is more prominent in new work; a negative bar means it is less prominent.

After 1980, new work became less concentrated in middle-paid production, transportation, construction, and clerical activities relative to existing work. For workers with more education, professional occupations became especially prominent. For workers with high school or less, personal and health services gained importance. Changes in the middle of the occupational distribution therefore involve both the decline of existing activities and a shift in where new activities emerge.

Exhibit 2. The changing location of new work



Historical evidence. Vertical axes are percentage-point differences in employment shares. The left panel covers workers with high school or less; the right panel is labelled some college or above. Red bars refer to 1940–1980 and blue bars to 1980–2018. Education is assigned using the workers observed in the occupation–industry cells containing the titles.

Question 1. What does new work measure? (20 marks)

(a) Interpret the 60% finding in Exhibit 1. Explain one reason it is an estimate rather than a direct count, and one reason a newly listed title need not describe work invented in that year. [10 marks]

(b) Use Exhibit 2 to evaluate the claim: "The changing occupational distribution after 1980 can be understood entirely as the disappearance of old middle-paid jobs." Compare the two education groups. [10 marks]

4 Two ways innovation changes work

An *automation innovation* substitutes for a worker's existing activities (and removes tasks). An *augmentation innovation* expands the capabilities, quality, variety, or usefulness of what an occupation produces (potentially creating new work). We can look at patent descriptions in search for information about new technologies and their interaction with old and new work.

To measure automation exposure, match the text of patents to descriptions of activities workers perform, such as sorting mail or processing records. To measure augmentation exposure, match patents to occupational and industry titles that describe outputs and specialties, such as nail treatment or solar electrical installations.

The idea is to match a patent for an automatic mail-processing apparatus to mail handlers as automation. A patent for strengthening and repairing fingernails is instead matched to personal-appearance services

as augmentation. The first substitutes for an input to an existing service. The second expands what a service can offer and can create demand for specialized workers. One invention can receive both kinds of match: a technology can replace some activities while opening up other work.

For each occupation and period, count the matched patents, giving more weight to patents that receive more citations. The logarithms of these weighted counts are the augmentation exposure A_{ot} and automation exposure M_{ot}. While patents are only measuring relevant inventions, rather than workplace adoption, we hope to see a correlation between these technological advancements and changes in the workplace captured by the emergence of new job titles (specialties).

Let n_{ot} be the number of new titles in occupation o during period t. The comparison can be written as

$$\log \mathrm{E}[n_{ot} \mid A_{ot}, M_{ot}, C_{ot}] = \beta_A A_{ot} + \beta_M M_{ot} + C'_{ot}\gamma. \tag{2}$$

Here C_{ot} contains the controls. A count regression allows an occupation to add no titles in a period. The coefficients are elasticities: they describe the proportional change in the expected new-title count associated with a proportional change in matched patent exposure. The controls include initial occupational employment shares and period effects; the final column also allows each broad occupational group to have its own period effect.

Exhibit 3. Which innovations predict new titles?

	(1) Augmentation	(2) Automation	(3) Both	(4) Both + group
$100\widehat{\beta}_A$	17.81	—	16.85	21.02
	(3.52)		(3.96)	(3.54)
$100\widehat{\beta}_M$	—	12.75	1.89	2.35
		(3.93)	(4.52)	(4.07)
Initial employment share	Yes	Yes	Yes	Yes
Period effects	Yes	Yes	Yes	Yes
Group-specific period effects	No	No	No	Yes

New-title count regressions, 1940–2018; 1,535 occupation–period observations. Coefficients and standard errors (parentheses) are multiplied by 100. Observations are weighted by initial employment.

In the final column, 10% greater augmentation exposure is associated with approximately 2.1% more new titles. The automation coefficient is close to zero relative to its standard error once augmentation exposure is included. The comparison changes substantially from the specification that includes automation alone.

Question 2. Separating the forces behind new work (25 marks)

(a) Explain why matching patents to an occupation's outputs can provide different information from matching them to its existing tasks. Use one of the examples in the case. Could the same invention contribute to both exposure measures? [10 marks]

(b) A commentator uses column (2) of Exhibit 3 to argue that automating tasks generates new human work. Evaluate that argument using the other columns. Explain the role of the group-specific period effects in column (4). [15 marks]

5 From associations to effects

The comparisons in Exhibit 3 need not isolate a causal effect. It might be that innovation does not generate new work titles. Instead, an expanding market can attract inventors while also encouraging employers to develop new specialties. In that case, innovation and new titles respond to a common demand change.

We need to find an instrument for the exposure of occupations to innovation that is uncorrelated with things like demand factors that can simultaneously affect innovation and work titles. One option is to use earlier *breakthrough patents* to predict later innovation. A breakthrough is both unusual relative to preceding patents and influential for subsequent patents. To find breakthrough patents we can rank all patents by how closely later inventions resemble them relative to their similarity to earlier inventions; the top tenth of patents within a decade supplies the breakthrough group. Breakthroughs in a technological field predict more patenting in that field about twenty years later, so they do not reflect those future demand conditions.

For each occupation, use its links to these earlier patents to construct separate augmentation and automation instruments, Z^A_{ot} and Z^M_{ot}. Each indicates whether the occupation's exposure to the corresponding breakthroughs twenty years earlier was above the median. Controls also account for exposure to *all* matched patents from the earlier period, so the instruments capture breakthrough exposure beyond ordinary prior patenting.

The first steps predict the two current exposures jointly:

$$A_{ot} = \pi_{AA} Z^A_{ot} + \pi_{AM} Z^M_{ot} + C'_{ot}\rho_A + v^A_{ot}, \tag{3}$$

$$M_{ot} = \pi_{MA} Z^A_{ot} + \pi_{MM} Z^M_{ot} + C'_{ot}\rho_M + v^M_{ot}. \tag{4}$$

For occupations with a positive new-title count, the second step is

$$\log n_{ot} = \beta_A \widehat{A}_{ot} + \beta_M \widehat{M}_{ot} + C'_{ot}\gamma + u_{ot}. \tag{5}$$

The hats denote the fitted exposures from the first steps. These regressions compare occupations whose later innovation differs because of their exposure to earlier breakthroughs, conditional on the controls. A causal interpretation requires that those breakthroughs affect later new-title creation through the measured innovation flows, rather than through an omitted force that independently changes demand

for the occupation. We can then repeat this process for employment growth within an occupation.

Exhibit 4. Instrumental-variable estimates for titles and employment

Outcome	$100\widehat{\beta}_A$	$100\widehat{\beta}_M$
Log new-title count	19.52	4.21
	(6.66)	(9.71)
Change in log employment, per decade	3.63	−4.21
	(0.96)	(0.93)
Change in log wage bill, per decade	4.04	−3.43
	(0.98)	(0.97)

Coefficients and standard errors are multiplied by 100. The first row uses 1,276 occupation–period observations with positive title counts and broad-group period effects. The last two rows use 33,900 occupation–industry changes over two forty-year intervals, with industry and broad-group period effects. All include both exposures, prior patent-exposure controls, and initial employment weights. Standard errors are clustered at the occupation-by-forty-year-period level for titles and the occupation–industry level for employment and wage bills.

A 10% increase in augmentation exposure corresponds to approximately 2.0% more new titles and 0.36% more employment growth per decade in the respective regressions. For automation, the new-title estimate remains statistically unclear, while the employment estimate implies approximately 0.42% less growth per decade. The wage bill is total labour earnings, so it can change because employment, hours, or wages change. Its increase does not by itself establish that the average worker's wage rose.

Cheatsheet: interpreting the empirical comparisons
A coefficient in a multiple regression holds the other included variables fixed. Fixed effects absorb differences common to the specified group and period. An approximate 95% interval is the estimate plus or minus twice its standard error. An instrumental variable must predict the exposure of interest and must not affect the outcome through an omitted alternative channel.

Question 3. Identification and employment (30 marks)

(a) Explain why current demand could confound the relationship between innovation and new work. How does the breakthrough-patent strategy try to address that problem? State a remaining threat to its causal interpretation. [15 marks]

(b) Interpret the difference between the new-title and employment results for automation in Exhibit 4. What does the augmentation wage-bill result establish about workers' earnings, and what does it leave unresolved? Explain why these estimates do not by themselves measure the net change in employment for the whole economy. [15 marks]

6 New work also responds to demand

An invention is not the only reason to create a new specialty. Consider the example given in the opening of this case study about personal care. As the economy grows and people become more affluent, the demand for personal care increases. Firms can then offer a wider range of services and workers can specialize in providing them. Conversely, if a manufacturing industry loses customers, employers may develop fewer new positions even when relevant technologies continue to improve. These examples capture the idea that the return to developing expertise and new specialties depends on the market for what it produces.

We can run additional regressions to explore this channel. The first uses rising Chinese import competition as a source of falling demand for U.S. production. Import growth in other high-income countries helps identify changes in Chinese supply, rather than relying only on U.S. imports that also respond to domestic demand. An occupation's exposure depends on the industries in which its workers were employed before the shock. The idea here is that some occupations are more exposed to import competition because they are predominantly in industries that end up facing more competition in the future.

The second uses population ageing to predict changes in demand. Age groups purchase different products and services. Combining their spending patterns with changes in population age structure predicts which industries face growing demand. Initial employment patterns then translate industry demand into occupational exposure. This channel can help explain new work in personal and care services that have relatively little exposure to patent-based augmentation.

Exhibit 5. Market demand and the creation of specialties

Comparison	$100\widehat{\beta}$	Standard error ×100
Import competition and new titles, 1990–2018	−17.49	5.13
Later import competition and earlier new titles, 1970–1990	−2.99	13.24
Demographic demand growth and new titles, 1980–2018	14.86	6.71

Dependent variable: new-title count, estimated using separate count regressions. Higher import exposure means more competition; higher demographic exposure means more predicted demand. All specifications control for augmentation exposure, initial occupational employment shares, period effects, broad occupational groups, and the sum of industry exposure weights. There are 610 observations in the first and third rows and 588 in the second. Standard errors are clustered by occupation. The exposure measures are different indices, so coefficient magnitudes are not directly comparable across the trade and demographic rows.

Greater import competition is associated with fewer new titles after 1990. It does not significantly predict title creation in the two preceding decades. That earlier-period comparison asks whether the exposed occupations were already on a different path before the trade shock. The demographic result points in the opposite direction: expanding demand supports the emergence of specialties, even after accounting for augmentation exposure.

Cheatsheet: connecting new work to the task model
Consider a unit-length range of tasks $[N-1, N]$, aggregated with Cobb–Douglas. Machines perform $[N-1, I]$ and workers perform $(I, N]$, with $N-1 < I < N$. Under competitive factor payments, equal expenditure across tasks gives the labour income share $s_L = N - I$. Automation advances I; new tasks performed by workers advance N. Employment also satisfies $L = Y\ell$, where $\ell = L/Y$ is labour required per unit of output. Thus $\Delta \log L = \Delta \log Y + \Delta \log \ell$.

Question 4. Demand, new tasks, and displacement (25 marks)

(a) Explain the purpose of the earlier-period comparison in Exhibit 5. What do the trade and demographic results add to an explanation of new work based only on inventions? [10 marks]

(b) A commentator argues: "Because new specialties keep appearing, new work must offset the employment effects of automation." Evaluate the claim using the task model and the evidence in this case. [15 marks]

Reference

Autor, David, Caroline Chin, Anna Salomons, and Bryan Seegmiller (2024). *New Frontiers: The Origins and Content of New Work, 1940–2018*. *Quarterly Journal of Economics* 139(3), 1399–1465.